

Pearson correlation coefficient

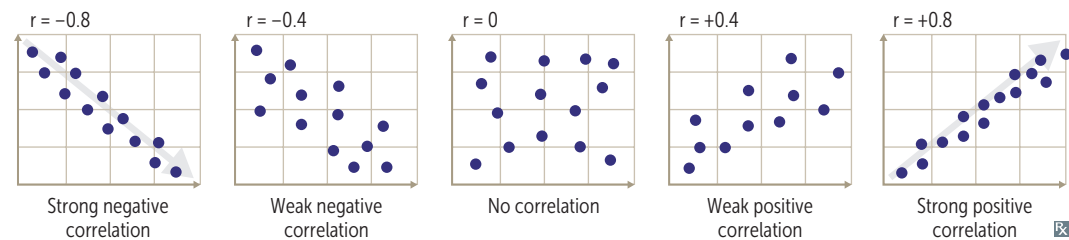
A measure of the linear correlation between two variables. r is always between -1 and $+1$. The closer the absolute value of r is to 1 , the stronger the linear correlation between the 2 variables.

Variance is how much the measured values differ from the average value in a data set.

Positive r value → positive correlation (as one variable ↑, the other variable ↑).

Negative r value → negative correlation (as one variable ↑, the other variable ↓).

Coefficient of determination = r^2 (proportion of variance in one variable that can be explained by variance in the other variable). Correlation does not necessarily imply causation.



► PUBLIC HEALTH SCIENCES—ETHICS

Core ethical principles

Autonomy	Obligation to respect patients as individuals (truth-telling, confidentiality), to create conditions necessary for autonomous choice (informed consent), and to honor their preference in accepting or not accepting medical care.
Beneficence	“Do good.” Physicians have a special ethical (fiduciary) duty to act in the patient’s best interest. May conflict with autonomy (an informed patient has the right to decide) or what is best for society (eg, mandatory TB treatment). Traditionally, patient interest supersedes. Principle of double effect—facilitating comfort is prioritized over potential side effects (eg, respiratory depression with opioid use) for patients receiving end-of-life care.
Nonmaleficence	“Do no harm.” Must be balanced against beneficence; if the benefits outweigh the risks, a patient may make an informed decision to proceed (most surgeries and medications fall into this category).
Justice	To treat persons fairly and equitably. This does not always imply equally (eg, triage).

Decision-making capacity

Physician must determine whether the patient is psychologically and legally capable of making a particular healthcare decision.

Note that decisions made with capacity cannot be revoked simply if the patient later loses capacity.

Intellectual disabilities and mental illnesses are not exclusion criteria unless the patient’s condition presently impairs their ability to make healthcare decisions.

Capacity is determined by a physician for a specific healthcare-related decision (eg, to refuse medical care).

Competency is determined by a judge and usually refers to more global categories of decision-making (eg, legally unable to make any healthcare-related decision).

Four major components of decision-making:

- Understanding (what do you know about your condition/proposed procedure/treatment?)
- Appreciation (what does your condition mean to you? why do you think your doctor is recommending this course of treatment?)
- Reasoning (how are you weighing your options?)
- Expressing a choice (what would you like to do?)

Informed consent

A process (not just a document/signature) that requires:

- Disclosure: discussion of pertinent information, including risks/benefits (using medical interpreter, if needed)
- Understanding: ability to comprehend
- Capacity: ability to reason and make one's own decisions (distinct from competence, a legal determination)
- Voluntariness: freedom from coercion and manipulation

Patients must have a comprehensive understanding of their diagnosis and the risks/benefits of proposed treatment and alternative options, including no treatment.

Patients must be informed of their right to revoke written consent at any time, even orally.

Exceptions to informed consent (**WIPE** it away):

- **Waiver**—patient explicitly relinquishes the right of informed consent
- Legally **Incompetent**—patient lacks decision-making capacity (obtain consent from legal surrogate)
- Therapeutic **Privilege**—withholding information when disclosure would severely harm the patient or undermine informed decision-making capacity
- **Emergency** situation—implied consent may apply

Consent for minors

A minor is generally any person < 18 years old. Parental consent laws in relation to healthcare vary by state. In general, parental consent should be obtained, but exceptions exist for emergency treatment (eg, blood transfusions) or if minor is legally emancipated (eg, married, self-supporting, or in the military).

Situations in which parental consent is usually not required:

- **Sex** (contraception, STIs, prenatal care—usually not abortion)
- **Drugs** (substance use disorder treatment)
- **Rock and roll** (emergency/trauma)

Physicians should always encourage healthy minor-guardian communication.

Physician should seek a minor's assent (agreement of someone unable to legally consent) even if their consent is not required.

Advance directives

Instructions given by a patient in anticipation of the need for a medical decision. Details vary per state law.

Oral advance directive

Incapacitated patient's prior oral statements commonly used as guide. Problems arise from variance in interpretation. If patient was informed, directive was specific, patient made a choice, and decision was repeated over time to multiple people, then the oral directive is more valid.

Written advance directive

Delineates specific healthcare interventions that patient anticipates accepting or rejecting during treatment for a critical or life-threatening illness. A living will is an example.

Medical power of attorney

Patient designates an agent to make medical decisions in the event that the patient loses decision-making capacity. Patient may also specify decisions in clinical situations. Can be revoked by patient if decision-making capacity is intact. More flexible than a living will.

Do not resuscitate order

DNR order prohibits cardiopulmonary resuscitation (CPR). Patient may still consider other life-sustaining measures (eg, intubation, feeding tube, chemotherapy).

Ventilator-assisted life support

Ideally, discussions with patients occur before ventilator support is necessary. However, information about patient preferences may be absent at the time patients require this intervention to survive. Medical decision-making frequently relies on surrogate decision-makers (patient identified or legally appointed) when discussing the continuation or withdrawal of ventilatory support, focusing on both the prognosis of the condition and the believed wishes of the patient.

If surrogates indicate patient would not have wanted to receive life support with ventilation → withhold or withdraw life support regardless of what the surrogate prefers.

If the decision is made to withhold or withdraw life support, involve palliative care, chaplain services, and the primary care physician in medical discussions with the family and provide emotional support.

Surrogate decision-maker

If a patient loses decision-making capacity and has not prepared an advance directive, individuals (surrogates) who know the patient must determine what the patient would have done. Priority of surrogates: **spouse** → adult **children** → **parents** → adult **siblings** → other relatives (the **spouse chips** in).

Confidentiality

Confidentiality respects patient privacy and autonomy. If the patient is incapacitated or the situation is emergent, disclosing information to family and friends should be guided by professional judgment of patient's best interest. The patient may voluntarily waive the right to confidentiality (eg, insurance company request).

General principles for exceptions to confidentiality:

- Potential physical harm to self or others is serious and imminent
- Alternative means to warn or protect those at risk is not possible
- Steps can be taken to prevent harm

Examples of exceptions to patient confidentiality (many are state specific) include the following ("The physician's good judgment **SAVED** the day"):

- Patients with active **Suicidal/homicidal** ideation
- **Abuse** (children, older adults, and/or prisoners)
- Duty to protect—state-specific laws that sometimes allow physician to inform or somehow protect potential **Victim** from harm
- Patients with **Epilepsy** and other impaired automobile drivers
- Reportable **Diseases** (eg, STIs, hepatitis, food poisoning); physicians may have a duty to warn public officials, who will then notify people at risk. Dangerous communicable diseases, such as TB or Ebola, may require involuntary treatment.

Accepting gifts from patients

A complex subject without definitive regulations. Some argue that the patient-physician relationship is strengthened through accepting a gift from a patient, while others argue that negative consequences outweigh the benefits of accepting any gift.

In practice, patients often present items such as cards, baked goods, and inexpensive gifts to physicians. The physician's decision to accept or decline is based on an individual assessment of whether or not the risk of harm outweighs the potential benefit.

- Physicians should not accept gifts that are inappropriately large or valuable.
- Gifts should not be accepted if the physician identifies that the gift could detrimentally affect patient care.
- Gifts that may cause emotional or financial stress for the patient should not be accepted.

If a gift violates any of the guidelines above, the best practice is to thank the patient for offering a kind gift, but politely indicate that it must be declined. During this conversation it should be emphasized that the incident does not influence the physician-patient relationship in any way.

► PUBLIC HEALTH SCIENCES—COMMUNICATION SKILLS

Patient-centered interviewing techniques

Introduction	Introduce yourself and ask the patient their name and how they would like to be addressed. Address the patient by the name and pronouns given. Avoid making gender assumptions. Sit at eye level, near the patient, while facing them directly.
Agenda setting	Identify concerns and set goals by developing joint agenda between the physician and the patient.
Reflection	Actively listen and synthesize information offered by the patient, particularly with respect to primary concern(s).
Validation	Legitimize or affirm the patient's perspectives.
Recapitulation	Summarize what the patient has said so far to ensure correct interpretation.
Facilitation	Encourage the patient to speak freely without guiding responses or leading questions. Allow the patient to ask questions throughout the encounter.

Establishing rapport**PEARLS**

Partnership	Work together with patient to identify primary concerns and develop preferred solutions.
Empathy	Acknowledge the emotions displayed and demonstrate understanding of why the patient is feeling that way.
Apology	Take personal responsibility when appropriate.
Respect	Commend the patient for coming in to discuss a problem, pushing through challenging circumstances, keeping a positive attitude, or other constructive behaviors.
Legitimization	Assure patient that emotional responses are understandable or common.
Support	Reassure patient that you will work together through difficult times and offer appropriate resources.

Delivering bad news**SPIKES**

Setting	Offer in advance for the patient to bring support. Eliminate distractions, ensure privacy, and sit down with the patient to talk.
Perception	Determine the patient's understanding and expectations of the situation.
Invitation	Obtain the patient's permission to disclose the news and what level of detail is desired.
Knowledge	Share the information in small pieces without medical jargon, allowing time to process. Assess the patient's understanding.
Emotions	Acknowledge the patient's emotions, and provide opportunity to express them. Listen and offer empathetic responses.
Strategy	If the patient feels ready, discuss treatment options and goals of care. Offer an agenda for the next appointment. Giving control to the patient may be empowering. Ask how they feel a problem might be solved and what they would like to do about the plan of action.